ROIP Quick Start Guide

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Five steps link two 551 ROIP modules over the internet, firmware v85. Do every step at both sites.

What you need

- **Two ROIP modules**, one per site, both on the same firmware.
- **At each site**: a free LAN address for the module, the netmask and gateway, and the site's public address. A static public IP or a dynamic DNS name both work.
- **Access to each site's router** to forward ports.
- **A way to configure the module**: a USB cable and a terminal at 115200 baud with local echo on, or a phone or laptop with Wi-Fi.
- **A link password** you choose. It must be identical on both modules.

Fill in this table before you start. The example values show the pattern:

Setting	Site A (example)	Site B (example)
Module LAN address	192.168.1.50	192.168.10.50
Netmask	255.255.255.0	255.255.255.0
Gateway (router LAN address)	192.168.1.1	192.168.10.1
This site's public address	203.0.113.10	siteb.mywire.org
Remote address to enter	siteb.mywire.org	203.0.113.10
Link password	same on both	same on both

Each module's remote address is the **other** site's public address. Tip: configure both modules on the bench before taking them to the sites.

Step 1: configure each module

Power the module and plug in Ethernet. Then use either method below; both ask the same questions.

Option A: USB console.

1. Connect USB and open the terminal at 115200 baud.
2. Type `//setup` and press Enter.
3. Answer each question from your table. Press Enter to keep the value in brackets:
   a. Static or DHCP: `S`
   b. Static IP address, netmask, gateway
   c. Remote device WAN address: the **other** site's public address
   d. Link password
   e. Audio input and output: `F` (flat) for speaker or line audio; `D` in and `P` out for a radio's discriminator and modulator (see Step 3)
   f. Codec: `PCM` on a fast connection; `OPUS` with bitrate `16000` on a slow one (about 30 kbps each way)
4. Answer `Y` to save and `Y` to reboot.

Option B: Wi-Fi from a phone.

1. Hold the mode button while powering up, and keep holding for 2 s.
2. Point the phone camera at the QR code on the display to join the module's Wi-Fi network, or join `SRS-ROIP-xxxx` by name with the password shown beside the code.
3. The setup page opens by itself, or browse to `http://192.168.4.1`.
4. Fill in the form, including Audio, and tap **Save and restart**.

If you choose DHCP instead of a static address, reserve the module's address in the router. The MAC address is printed at start-up and by `//net`.

Step 2: forward ports on each router

On each site's router, forward these four ports to that site's module LAN address:

Protocol	Port	Carries
UDP	7300	Audio
UDP	7301	Heartbeat
UDP	7302	GPIO
TCP	7303	Serial (DCN)

Port 7303 is **TCP**, not UDP; a UDP-only rule is the most common reason the serial channel won't connect. If the router allows it, limit each rule to the other site's public address.

A site whose router WAN address is 100.64.x.x to 100.127.x.x is behind carrier-grade NAT and can't accept incoming traffic; ask the ISP for a public address or use a VPN.

Step 3: connect the radio and set emphasis

Wire the radio or controller to J3: audio in and out, **PTT_INPUT** (pin 4, pull low to send) and **COR_OUTPUT** (pin 2, goes low while audio arrives from the far site). Then check the emphasis you chose in Step 1 matches what you wired, or set it here:

Wired to	Type these commands
Speaker or line audio (flat)	Nothing: both paths are flat by default
A radio's discriminator and modulator, or a controller that expects them	`//audioin,de` `//audioout,pre` `//rxhpf,on` `//txlpf,on`

Pre-emphasis lowers the output level at 1 kHz by about 9 dB; Step 5 sets it back.

Optional: on a slow internet connection, `//codec,opus` with `//opusbitrate,16000` cuts each direction to about 30 kbps with very good voice (both modules on v85). For a DCN bus, set its baud rate with `//dcnbaud,9600` (your bus speed) on both modules.

Step 4: check the link

With both modules running, type `//status` on each. Every line should look like this:

```
> //status
--- Status ---
  My IP address      192.168.1.50  (static)
  Ethernet           connected
  Remote IP address  siteb.mywire.org  (198.51.100.20)
  Remote device      connected
  Audio channel      working
  Serial channel     working
  GPIO channel       working
--------------
```

The Status and Link LEDs are on solid at both sites when the pair is linked.

The quickest check is `//linktest` on each module. It tests Ethernet, the gateway, the internet, the remote's address and each stream in each direction, and prints PASS or FAIL with what to fix:

```
> //linktest
--- Link test (takes about 10 s; audio keeps working) ---
  [PASS] Ethernet link             up
  ...
  [PASS] Serial (TCP 7303)         connected, we dialed out
--- 10 passed, 0 failed, 0 warning(s) ---
Everything checked is working.
```

Without a laptop, press the mode button until the **LINK CHECK** page appears: a tick for every stream in both directions and "All working" mean the pair is linked.

If `//linktest` shows a FAIL, do the fix on its line; the first FAIL is the one to fix first. The common ones, as //status words them:

Line shows	Fix
`Ethernet ... gateway doesn't answer`	This module's IP, netmask or gateway is wrong: run `//setup` again
`Remote device NOT connected`	Check the remote address is the other site's public address, and that the UDP ports are forwarded at both sites
Audio or GPIO `NOT working`	One router is missing that port: `//status,detail` names the port and the site
`Serial channel NOT working`	Forward **TCP** 7303 at one site at least, and check the link passwords match; then `//connect`

After fixing a router, `//connect` retries the serial link at once instead of waiting.

Step 5: set audio levels

1. At site A, send a 10-second 1 kHz test tone to site B:

   ```
   //tone,1000,10000
   ```

2. At site B, adjust what comes out with `//txlevel,<0-100>` until the radio's deviation or the controller's input level is right. 50 is unity; each step above 50 adds 0.64 dB. With `//audioout,pre` set, start at about 64.
3. Key PTT at site A with real audio and check it at site B. If it's too quiet or too loud going in, adjust `//rxlevel,<0-100>` at site A.
4. Repeat in the other direction.

To set levels by numbers instead of by ear, show the **LEVELS** page or type `//levels`: it gives the peak level of the audio going in and coming out, in dBFS. Aim for peaks of -6 to -3 dBFS on normal speech, and never CLIP. Input is too hot: lower `//rxlevel`. Output is clipping: lower `//rxlevel` at the other site.

All settings are saved and survive a power cycle. `//net` lists every setting, and the full ROIP Module User Guide covers each command. If you need help, `//report` prints everything support needs in one block, with the password hidden.